

Bluetooth Decryption

Privacy Analysis of Augmented Reality Glasses

1 Objective

To decrypt Bluetooth traffic captured between the POCO X5 5G phone and Ray-Ban Meta glasses using tools such as Wireshark, btsnoop_hci.log, and decryption keys extracted from bt_config.conf.

2 Steps

Note that the following will require root access otherwise the locations will not be visible.

1. Make sure to be a Developer on the phone.
2. Go to **Settings, Developer Options**
3. Enable **Bluetooth HCI Snoop log**
4. Wait for one minute
5. Pair and interact with the phone
6. Wait for one minute
7. Generate a bugreport using the command:

```
adb bugreport bugreport.zip
```

8. Extract the zip file and look for the snoop log inside. It should be in /bugreport/FS/data/misc/bluetooth/logs/.
If you do not see the snoop log, wait for a couple minutes and then try again.
9. Next, we need to locate the **bt_config.conf** file, which should be in /data/misc/bluedroid/.
You can for its existence by using the following commands:

```
adb shell
su
cd /data/misc/bluedroid/
ls
```

If you see the file **bt_config.conf**, pull it onto the laptop by opening a new terminal and using the command:

```
adb pull /data/misc/bluedroid/bt_config.conf
```

Or the command, if there are permissions issues:

```
adb shell "su -c 'cat /data/misc/bluedroid/bt_config.conf'" > bt_config.conf
```

If not, try a different path, e.g. /data/misc/bluetooth/.

10. Open the file using nano or gedit, or any preferred editor:

```
gedit bt_config.conf
```

11. Find the necessary keys for decryption, for me the important keys included:

- LinkKey for Classic Bluetooth
The encryption key used for secure connections.
- LE_KEY_PENC
Peripheral Encryption Key)for BLE
- LE_KEY_LENC
Local Encryption Key for BLE
- IRK
Identity Resolving Key to resolve random addresses.
- LE_KEY_PCSRK/LE_KEY_LCSRK
Signature resolving keys (used for data signing).

12. Edit the preference file to add the keys manually:

```
nano ~/.config/wireshark/preferences
```

13. Add the following into the file, (the mac address of the glasses plus the key obtained from the .conf file):

```
bluetooth.br_edr_link_key:98:59:49:34:7a:8c,...  
bluetooth.le_ltk:98:59:49:34:7a:8c,...  
bluetooth.le_irk:98:59:49:34:7a:8c,...
```

14. Load the **btsnoop_hci.log** on Wireshark. It is seen that Wireshark successfully parsed the HCI packets (bthci_cmd and bthci_evt), but higher-layer protocols (e.g., btcommon) were not visible. Also BLE Attribute Protocol (ATT) traffic were visible (btatt) but with encrypted payloads.

3 Problems Encountered

- The packets in Wireshark remained encrypted despite adding the keys to preferences.
- The traffic showed random MAC addresses (e.g., 48:2f:2e:1b:12:b0) instead of the glasses' static address (98:59:49:34:7a:8c). One can resolve this by adding the IRK to resolve the address. However, no improvements were observed.
- Permission denied when trying to copy the **bt_config.conf** onto the laptop.
This is because the file is part of the system's protected directories, and accessing it requires elevated privileges, even with a rooted device. One can resolve this by pulling the file with elevated privileges:

```
adb shell "su -c 'cat /data/misc/bluedroid/bt_config.conf'" > bt_config.conf
```

- Wireshark UI cannot add keys.
It is alleged that on Wireshark, one can directly add the keys with the following steps:
 1. Open Wireshark and go to **Edit, Preferences**.
 2. Go to **Protocols**, then scroll down to **Bluetooth**.
 3. Add the keys under the decryption option.

One can try to resolve this by rechecking the Wireshark installation.

- Attempted to convert btsnoop_hci.log to PCAP for compatibility by using the btsnoop-to-pcap script. However, dependencies and incomplete scripts blocked progress.

4 Additional Notes

For the keys: If You Don't Re-Pair the glasses, the keys (like the LinkKey) in `bt_config.conf` usually remain the same for the same paired connection, unless you unpair and repair the device. So, if you're just capturing new sessions without unpairing the glasses, the file will not need to be pulled again. But, if you unpair and re-pair the glasses, or factory reset the glasses or phone, the encryption keys will likely change, and you'll need to pull the updated `bt_config.conf` file.

However, for LE keys, they may change for each session if the pairing process happens again or if a session-specific key is negotiated (which is common in BLE). If decryption fails after recapturing traffic, the updated file may need to be pulled.

One can reset the bluetooth HCI Snoop log by:

1. Go to **Settings Developer Options**.
2. Turn Enable Bluetooth HCI Snoop Log off.
3. Turn off Bluetooth.
4. Turn it back on after a few seconds.
5. Re-enable Enable Bluetooth HCI Snoop Log.

This will start capturing logs for the new Bluetooth session without requiring a reboot or file deletion. Note, to wait a minute before starting the interaction and pulling the log.